Anant Maheshwary

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.58
Pace Junior Science College, Borivali	Class XII (Maharashtra State Board of Secondary & Higher Secondary Education)	2024	95.17%
Children's Academy, Mumbai	Class X (Council for the Indian School Certificate Examinations)	2022	99.20%

SCHOLASTIC ACHIEVEMENTS

Academic Excellence Award: Highest CGPA in B.Tech Computer Science and Engineering for the Year 2025

NSEP, NSEA: Placed in the National Top 1% in the National Standard Examinations in Physics and Astronomy, 2023

Gold Medal in National Assessment for Scientific Temperament and Aptitude (NASTA), 2021

SKILLS

Programming Languages: C/C++, Java, Python, Assembly(RISC-V), Bash, SQL, JavaScript, OCaml, HTML/CSS

Tools and Libraries: Git, Docker, eBPF, Sched_Ext, Flask, PyTest, NumPy, Matplotlib, Pandas

RELEVANT COURSES

Core CS: Computer Architecture, Database Management Systems, Operating Systems, Compilers, Software Development Fundamentals, Data Structures, Algorithms, Theory of Computation, Discrete Mathematics

Mathematics: Probability and Random Variables, Statistics, Linear Algebra, Transform Techniques

PROJECTS

Computer Architecture: RISC-V Pipelined Simulator *(C++, Python, RISC-V)*

- Built an interactive 5-stage pipelined RISC-V simulator with configurable hazard detection/resolution and branch prediction modes, supporting breakpoints, step execution and cycle-level debugging.
- Designed a modular OOP architecture that models hardware components independently, allowing new pipeline features and prediction strategies to be added with minimal changes.
- Developed an automated pytest-based verification framework and multi-stage testing suite for comprehensive testing across all simulator configurations.

SCX CARRY: IPC-Aware Hybrid CPU Scheduler (*C, eBPF, sched_ext)*

- Designed and implemented an IPC-aware Linux CPU scheduler using eBPF and the sched_ext framework, dynamically classifying tasks and scheduling them across Performance and Efficiency cores on hybrid CPUs.
- Developed a novel scheduling architecture with per-task runtime profiling, tiered dispatch queues, cache-affinity optimization and penalty-based heuristics to prioritize latency-critical interactive workloads.
- Benchmarked against Linux EEVDF and production schedulers using synthetic and real-world gaming workloads.

- Improved gaming responsiveness with consistently tighter frametimes and stronger low-percentile FPS, while maintaining comparable power efficiency to the default Linux scheduler.

xv6 Operating System (*C)*

- Extended xv6 with MLFQ scheduling, page replacement algorithms and RAID-based storage with disk scheduling policies.

Software Development: (*OCaml*)

- Extended an eager, statically typed MiniML compiler in OCaml by adding division and exception handling support across the compiler pipeline using OCamllex and Menhir.

Digital Circuits KMap Solver: (*Backend: Java, Python; Frontend: Javascript, HTML/CSS)*

- Built a full-stack web platform for Boolean expression minimization using the Quine-McCluskey algorithm, with circuit diagram, timing waveforms and Verilog code generation.

Arbitrary Precision Arithmetic Library *(Java, Maven, Docker)*

- Developed an arbitrary-precision integer and floating-point arithmetic library, packaged with Maven and containerized using Docker.

Database Management System: Query Optimizer Analysis (*PostgreSQL, Python)*

- Analyzed PostgreSQL query optimizer behavior across parameterized TPC-H queries by evaluating index selection, plan switches and forcing alternative execution plans using *pg_hint_plan*.

POSITIONS OF RESPONSIBILITY

Internship Coordinator, Office of Career Services, IITH: Facilitating internship opportunities by reaching out to companies and working in a team to resolve student and recruiter queries.

EXTRACURRICULAR

Competitive Programming: Codeforces: dwbh21 (Peak Rating: 1644 Expert)

Chess: (International FIDE Rating - 1577) Participated in Inter-IIT Sports Meet 2025

Sudoku and Cryptic Crossword Enthusiast